

1) Quadratic constraint

$$\left\| \frac{1}{2}(b^T x + c) \right\|_2 \leq \frac{1 - (b^T x + c)}{2}$$

$$b^T x + c = 0 \Rightarrow \sqrt{\left(\frac{1+B}{2}\right)^2 + \|Ax\|_2^2} \leq \frac{1-B}{2}$$

$$\Rightarrow \frac{(1+B)^2}{4} + \|Ax\|_2^2 \leq \frac{(1-B)^2}{4}$$

$$\Rightarrow \frac{(1+B)^2 - (1-B)^2}{4} + \|Ax\|_2^2 \leq 0$$

$$\Rightarrow \frac{B^2 + 2B + 1 - B^2 + 2B - 1}{4} + \|Ax\|_2^2 \leq 0$$

$$\Rightarrow B + \|Ax\|_2^2 \leq 0 \Rightarrow x^T A^T A x + b^T x + c \leq 0$$

2) QCBP $\rightarrow$ SOCP

$$\text{minimize } x_1^2 + 4x_2x_3 + 4x_3^2$$

$$\text{subject to } 9x_1^2 + 16x_2^2 \leq 25$$

$$x_1 - x_2 = 1$$

$\Rightarrow$ SOCP format: $\min^T v$

$$s.t. \rightarrow \|Ax + b\|_2 \leq c^T v + d$$

$$Ax = b$$

$$t = x_1^2 + 4x_2x_3 + 4x_3^2$$

$\Rightarrow$ minimize +

$$x_1^2 + 4x_2x_3 + 4x_3^2 \leq t \quad (1)$$

$$9x_1^2 + 16x_2^2 \leq 25 \quad (2)$$

$$x_1 - x_2 = 1$$

$$(1) \Rightarrow (x_1 + 2x_3)^2 \leq t$$

$$u^L \leq t \Leftrightarrow \left\| \frac{2u}{t-1} \right\|_2 \leq t+1$$

$$\Rightarrow \left\| \frac{2x_1 + 4x_3}{t-1} \right\|_2 \leq t+1$$

$$(2) \Rightarrow (3x_1)^2 + (4x_2)^2 \leq (5)^2$$

$$\Rightarrow \|3x_1 + 4x_2\|_2^2 \leq 5^2$$

$$\Rightarrow \|3x_1 + 4x_2\|_2 \leq 5$$

$\Rightarrow$ minimize +

$$\text{subject to } \left\| \frac{2x_1 + 4x_3}{t-1} \right\|_2 \leq t+1$$

$$\|3x_1 + 4x_2\|_2 \leq 5$$

$$x_1 - x_2 = 1$$

Linear programming:

$$\text{minimize } \|(2x_1 + 3x_2, -3x_1)^T\|_\infty$$

$$\text{subject to } |x_1 - 2x_2| \leq 3$$

$\Rightarrow$ LP format: minimize $c^T x$

$$\text{subject to } Gx \leq h$$

$$Ax = b$$

$$\Rightarrow \text{minimize } \max\{2x_1 + 3x_2, 1 - 3x_1\}$$

"

$\Rightarrow$ minimize +

$$2x_1 + 3x_2 \leq t$$

$$1 - 3x_1 \leq t$$

"

$\Rightarrow$ minimize +

$$2x_1 + 3x_2 \leq t$$

$$2x_1 + 3x_2 \geq -t$$

$$-3x_1 \leq t$$

$$-3x_1 \geq -t$$

$$x_1 - 2x_2 \leq 3$$

$$x_1 - 2x_2 \geq -3$$

unequal constraints

$$\Rightarrow 2x_1 + 3x_2 - t \leq 0$$

$$-2x_1 - 3x_2 - t \leq 0$$

$$-3x_1 - t \leq 0$$

$$3x_1 - t \leq 0$$

$$x_1 - 2x_2 - 3 \leq 0$$

$$-x_1 + 2x_2 - 3 \leq 0$$

$$2x_1 + 3x_2 - t + s_1 = 0 \quad s_1 \geq 0$$

$$-2x_1 - 3x_2 - t + s_2 = 0 \quad s_2 \geq 0$$

$$-3x_1 - t + s_3 = 0 \quad s_3 \geq 0$$

$$3x_1 - t + s_4 = 0 \quad s_4 \geq 0$$

$$x_1 - 2x_2 - 3 + s_5 = 0 \quad s_5 \geq 0$$

$$-x_1 + 2x_2 - 3 + s_6 = 0 \quad s_6 \geq 0$$

$$x_1 = x_1^+ - x_1^- \quad x_1^+, x_1^- \geq 0 \Rightarrow 2(x_1^+ - x_1^-) + 3(x_2^+ - x_2^-) - t + s_1 = 0$$

$$x_2 = "$$

$$s_2 = "$$

$$"$$

$$"$$

$$"$$

$$"$$

Transform LP $\rightarrow$ standard form

$$\text{minimize } c^T x + d$$

$$\text{subject to } Gx \leq h$$

$$Ax = b$$

$$\Rightarrow \text{minimize } p^T x'$$

$$\text{subject to } Bx' = e$$

$$x' \geq 0$$

a) minimize $c^T x + d$

$$Gx - h + s = 0 \quad s \geq 0$$

$$Ax = b$$

$$x = x^+ - x^- \quad x^+ \geq 0, x^- \geq 0$$

$$\Rightarrow \text{minimize } c^T(x^+ - x^-) + d \quad (1)$$

$$G(x^+ - x^-) + s = h \quad (2)$$

$$A(x^+ - x^-) = b \quad (3)$$

$$s \geq 0; x^+ \geq 0, x^- \geq 0$$

$$(2), (3) \Rightarrow \begin{bmatrix} G & -G & I \\ A & -A & 0 \end{bmatrix} \begin{bmatrix} x^+ \\ x^- \\ s \end{bmatrix} = \begin{bmatrix} h \\ b \end{bmatrix}$$

$$B$$

$$x'$$

$$e$$

$$(1) \quad \begin{bmatrix} c^T & -c^T & 0 \end{bmatrix} \begin{bmatrix} x^+ \\ x^- \\ s \end{bmatrix}$$

$$p^T$$

$$\Rightarrow \text{minimize } p^T x'$$

$$Bx' = e$$

$$x' \geq 0$$

explain: Feasible set $x = x^+ - x^-$ (x^+, x^-, s)

Optimal solution $x^* = x^{+*} - x^{-*}$ (x^{+*}, x^{-*}, s^*)

Optimal value: $f_{LP} c^T x + d$